

Program : Diploma in Renewable Energy	
Course Code : 2339	Course Title: Introduction to Computation and Programming Lab
Semester : 2	Credits: No credit
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To understand the fundamental concept of C programming and use it in problem solving.
- To develop logic that will help students to adapt to any other programming language.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic functions and features of Computer, Operating system and Internet applications, basic programming skills in Python.		Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Make use of various data types, I/O statements ,basic operators and control structures in C programming	13	Applying
CO2	Construct C programs based on arrays and string handling functions.	11	Applying
CO3	Develop C programs using user-defined functions.	9	Applying
CO4	Apply pointers and structure concepts in C programs .	9	Applying
	Series Test	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3	3			3
CO2	3	3	3	3		3	
CO3	3	3	3	3	3		
CO4	3	3	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Make use of various data types, I/O statements and basic operators in C programming		
M1.01	Demonstrate output and input functions.	2	Understanding
M1.02	Demonstrate different fundamental data types in C.	2	Understanding
M1.03	Solve simple problem statements using operators.	2	Applying
M1.04	Solve problem statements in C using different control structures(sequential, selection and iteration).	7	Applying
CO2	Construct C programs based on arrays and string handling functions.		
M2.01	Make use of one dimensional array operations - largest, smallest, compute sum & average of array elements, Search an array element using linear and binary search.	5	Applying
M2.02	Write programs to perform two dimensional array operations - matrix addition.	3	Applying
M1.03	Implement string manipulation functions - strcpy(), strcat(), strlen(), strcmp().	3	Applying
	Series Test 1	1.5	
CO3	Develop C programs using user-defined functions.		
M3.01	Illustrate 'User-defined functions'	6	Applying

M3.02	Write programs using recursive functions- Factorial of a number using recursion.	3	Applying
CO4	Apply pointers and structure concepts in C programs		
M4.01	Write programs to perform pointer operations- swapping of two numbers using pointers	4	Applying
M4.02	Illustrate structure concept -Display employee details and prepare rank list of students.	5	Applying
	Series Test 2	1.5	

**** - Suggested Open Ended Projects**

(Students have to do open ended experiments during the course for the purpose of continuous evaluation. This experiment shall be included in the bona-fide record.)

Example: Develop program such as

- Student CGPA calculator,
- Consumer billing,
- Candidate list sorting

Text / Reference:

T/R	Book Title/Author
T1	Let Us C – Yashavant Kanetkar, BPB Publications
R2	Programming in ANSI C, E. Balagurusamy, Tata McGraw Hill
R3	A Text Book on C, E. Karthikeyan, PHI Learning Pvt. Ltd
R4	Programming in C D. Ravichandran, New Age International Publishers
R5	C. Programming Language, Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall
R6	Programming in C, S. Jose, 1st Ed.
R7	Programming in C, Asok N Kamthane. Pearson.

Online Resources:

Sl.No	Website Link
1	http://freecomputerbooks.com/The-C-Programming-Language.html
2	https://www.tutorialspoint.com/cprogramming/index.htm

3	https://www.guru99.com/c-programming-tutorial.html
4	https://www.cprogramming.com/
5	http://www.zentut.com/c-tutorial/

Sample Questions To Test Outcomes

1. Find the sum of digits of a number.
2. Implement a C program to find the sum and average of all elements in an integer array of size MxN where M,N and array elements can be input while the program is executing.
3. Implement a program to check whether the given string is a Palindrome or not.
4. Write a program to accept a string and count the number of vowels present in this string
5. Program to display the rank list of students using structure concepts.